

Tiam Bennett Morrow-Rogers

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Education

University of Ottawa – BSc in Electrical Engineering

Expected May 2029

Humber College – BID Industrial Design

May 2022

Experience

Project Manager, University of Ottawa – Ottawa, ON

September 2025 – June 2026

- Trained and mentored 40+ students across two laboratories in embedded system programming (Arduino), hardware testing methodologies, and CAD modelling to elevate technical fabrication quality.
- Evaluated student hardware and software concepts weekly and provided structured technical feedback to ensure functional design execution, lab safety, and testability.

Product Manager and Designer, Phase Studio – Toronto, ON

January 2019 – July 2022

- Debugged and resolved firmware and hardware defects in C, KiCAD, and SolidWorks for 400+ deployed units.
- Managed end-to-end hardware development and overseas manufacturing (DFM/DFA) in China, establishing PCB quality testing standards to drive \$225k+ in single-month revenue.

Technical Communicator, APMA – Toronto, ON

April 2022 – July 2022

- Accelerated technical visualization for CES 2023 product demonstrations by modeling high-fidelity 3D assets in SolidWorks and Fusion 360.
- Translated complex technical specifications between international engineering teams and the CTO to align UI/UX prototype deliverables under strict exhibition deadlines.

Design Researcher, Humber College – Toronto, ON

November 2021 – May 2022

- Assessed manufacturing feasibility and mechanical integrity for physical product concepts by building 3D CAD models and rapid functional prototypes.
- Coordinated cross-faculty project deliverables by leading technical design workshops and user-centered physical prototyping workflows.

Projects

CANDU Pressure Tube Sampler | 1st Place Lifecycle Analysis

www.tiamdesigns.com/sample-subway

- Designed and built a custom pipe crawler using 3D printing (DFM) and embedded hardware, integrating an ESP32, HX711 load cell amplifier, battery management system (BMS), and H-bridge motor drivers.
- Developed Arduino/C++ firmware to read load cell data and drive motion controls for precise 30–80 mg nuclear pipe material sampling.
- Built a custom Python script on host PC to establish Wi-Fi and Bluetooth (Serial/Sockets) telemetry, enabling real-time remote motor control and sensor data acquisition.

Titan65 Custom Keyboard

www.tiamdesigns.com/Titan65

- Diagnosed and resolved mid-production PCB electrical defects, implementing hardware and firmware corrections that mitigated 65% of manufacturing triage costs.
- Executed manufacturing verification protocols with offshore suppliers and redesigned enclosure mounting components to eliminate mechanical clearance issues.

Skills

Languages: C, C++, SystemVerilog, Python, MATLAB, Java, JavaScript

EDA & Hardware: Xilinx Vivado, Quartus, KiCAD, Simulink, Oscilloscopes, Logic Analyzers, Git, Multimeters

CAD & Prototyping: SolidWorks, Fusion 360, Onshape, AutoCAD, Revit, 3D Printing (DFM)